중적분 14

연습문제 풀이

연습문제 14.1

1. 다음 반복적분을 구하여라.

(2) $\int_0^1 \int_0^1 e^{x+y} dy\, dx$

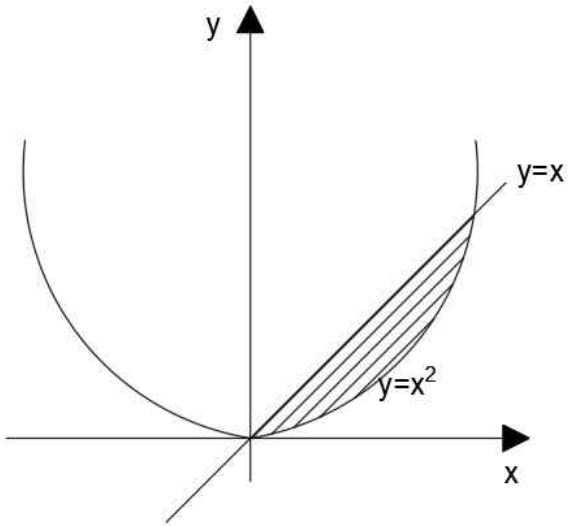
$$= \int_0^1 \left[e^{x+y}\right]_0^1 dx = \int_0^1 (e^{x+1} - e^x) dx$$

$$= \left[e^{x+1} - e^x\right]_0^1 = e^2 - e^1 - (e^1 - e^0) = e^2 - 2e + 1$$

$$(3)\ \int_0^1 \int_x^{x^2} 1\, dy\, dx$$

$$= \int_0^1 [y]_x^{x^2} dx = \int_0^1 (x^2 - x)dx$$

$$= \left[\frac{1}{3}x^3 - \frac{1}{2}x^2\right]_0^1 = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$$



*실제 적분 구간이 $\int_{x^2}^{x}$ 가 되어야 하나, 문제에선 $\int_x^{x^2}$ 으로 주어졌기 때문에, (−) 결과가 나오게 됨.

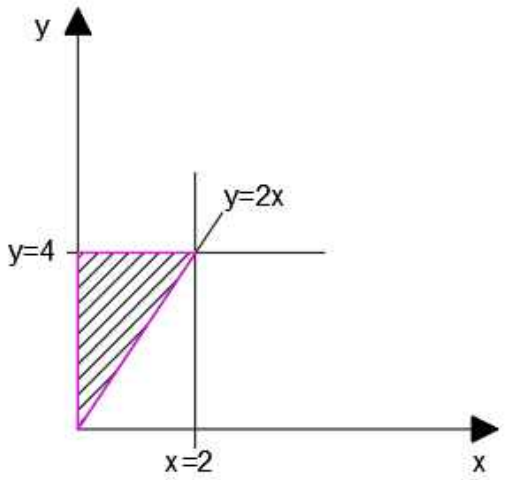
4. 다음 2중적분을 계산하여라.

(2) $\iint_R (x+y)dA$, R은 직선 $y=2x$, $x=0$, $y=4$에 의해 둘러싸인 영역

$$\int_0^2\int_{2x}^4 (x+y)dy\,dx = \int_0^2 \left[xy+\frac{1}{2}y^2\right]_{2x}^4 dx$$

$$= \int_0^2 \{(4x+8)-(2x^2+2x^2)\}dx = \int_0^2 (-4x^2+4x+8)dx$$

$$= \left[-\frac{4}{3}x^3+2x^2+8x\right]_0^2 = (-\frac{32}{3}+8+16)-0 = \frac{40}{3}$$



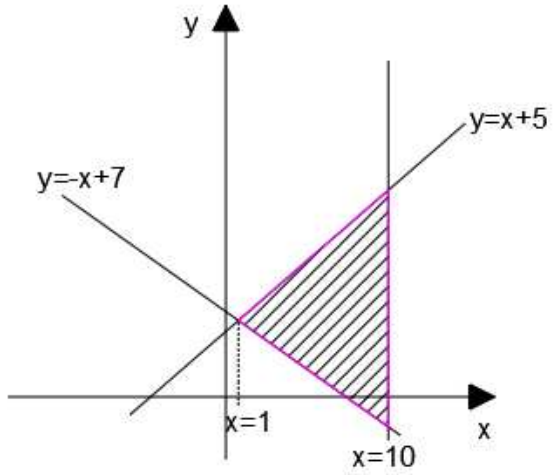
(4) $\int_R\int (3x-5)dA$, R은 직선 $y=x+5$, $y=-x+7$, $x=10$에 의해 둘러싸인 영역

$$\int_1^{10}\int_{-x+7}^{x+5}(3x-5)dy\,dx = \int_1^{10}[3xy-5y]_{-x+7}^{x+5}dx$$

$$=\int_1^{10}[\{3x(x+5)-5(x+5)\}-\{3x(-x+7)-5(-x+7)\}]dx$$

$$=\int_1^{10}(6x^2-16x+10)dx=[2x^3-8x^2+10x]_1^{10}$$

$$=2(10)^3-8(10)^2+10(10)-(2-8+10)=1296$$

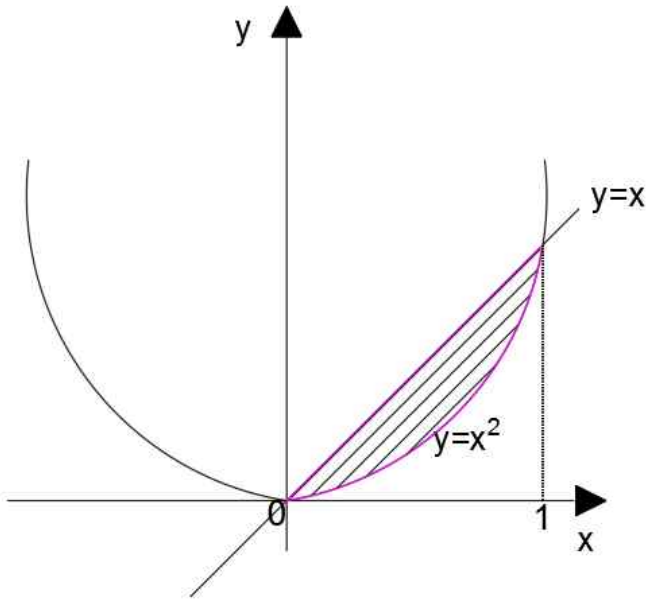


(5) $\int_R\int xy\,dA$, R은 직선 $y=x$, $y=x^2$의 그래프에 의해 둘러싸인 영역

$$\int_0^1\int_{x^2}^{x} xy\,dy\,dx = \int_0^1\left[\frac{1}{2}xy^2\right]_{x^2}^{x}dx$$

$$= \int_0^1\left(\frac{1}{2}x^3-\frac{1}{2}x^5\right)dx = \frac{1}{2}\left[\frac{1}{4}x^4-\frac{1}{6}x^6\right]_0^1$$

$$= \frac{1}{2}\left\{\frac{1}{4}-\frac{1}{6}\right\} = \frac{1}{24}$$

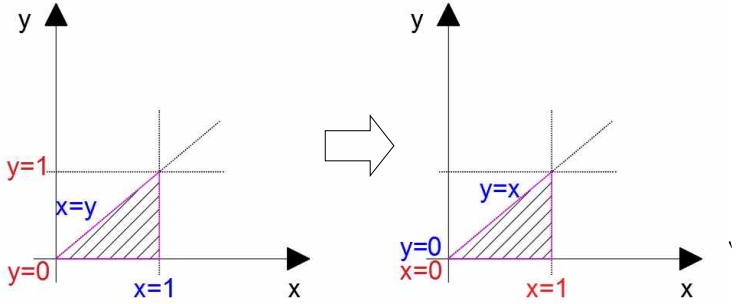


6. 적분의 순서를 바꾸어서 다음의 적분을 계산하여라.

(1) $\int_0^1 \int_y^1 e^{x^2} dx\, dy$

$\int_R\int e^{x^2} dx\, dy$, R은 $0 \le y \le 1$위의 $x = y$와 $x = 1$의 수평적 단일영역을

수직적 단일영역으로 바꾸면 (그래프 참조)



$$\int_0^1 \int_0^x e^{x^2} dydx = \int_0^1 \left[e^{x^2} y\right]_0^x dx = \int_0^1 e^{x^2} xdx$$

$x^2 = t$로 치환, $2xdx = 1dt \Rightarrow xdx = \frac{1}{2}dt$

$x = 0 \Rightarrow t = 0$, $x = 1 \Rightarrow t = 1$

$$= \int_0^1 e^t \frac{1}{2} dt = \frac{1}{2}\left[e^t\right]_0^1 = \frac{1}{2}(e-1)$$

(4) $\int_1^e \int_0^{\ln x} y\,dy\,dx$

$\int_R\int y\,dy\,dx$, R은 $0 \le y \le \ln x$위의 $x=1$와 $x=e$의 수직적 단일영역을

수평적 단일영역으로 바꾸면, (그래프 참조)

$$\int_0^1 \int_{e^y}^e y\,dx\,dy = \int_0^1 [yx]_{e^y}^e dy = \int_0^1 (ey - e^y y)\,dy$$

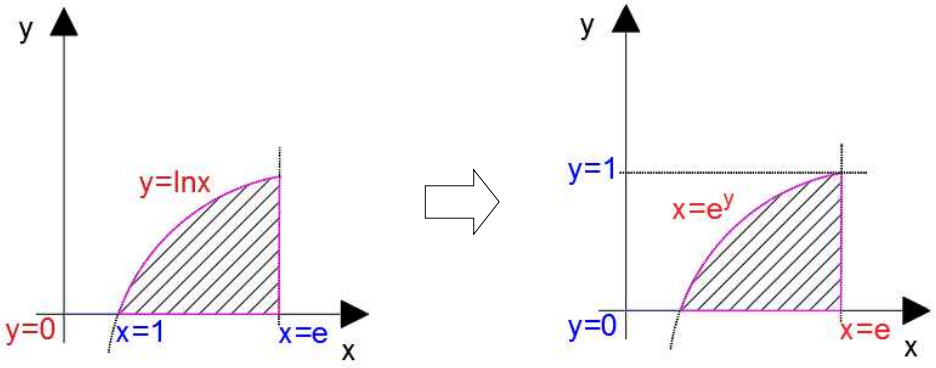
$$= \left[\frac{1}{2}ey^2\right]_0^1 - \int_0^1 ye^y\,dy$$

$u = y,\ v' = e^y$

$u' = 1,\ v = e^y$

$$= \frac{1}{2}e - \left\{[ye^y]_0^1 - \int_0^1 e^y\,dy\right\} = \frac{1}{2}e - e + [e^y]_0^1$$

$$= \frac{1}{2}e - e + e - 1 = \frac{1}{2}e - 1$$



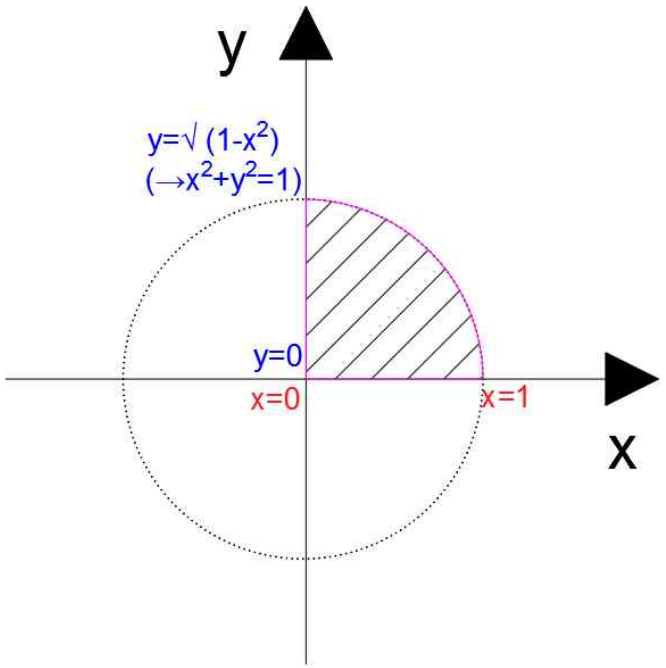
연습문제 14.2

3. 다음 적분을 극좌표 위의 반복적분으로 바꾸고 적분값을 계산하여라.

(1) $\int_0^1 \int_0^{\sqrt{1-x^2}} 1 dy dx$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 r dr d\theta = \int_0^{\frac{\pi}{2}} \left[\frac{1}{2} r^2 \right]_0^1 d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} d\theta = \left[\frac{1}{2} \theta \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$



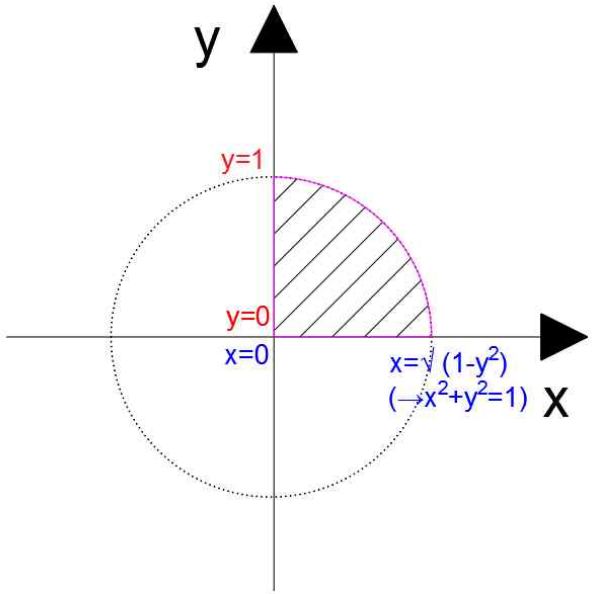
(2) $\int_0^1 \int_0^{\sqrt{1-y^2}} \sin(x^2+y^2)\,dxdy$

$$f(x,y)=\sin(x^2+y^2)$$
$$f(rcos\theta, rsin\theta)=\sin(r^2\cos^2\theta+r^2\sin^2\theta)=\sin(r^2(\cos^2\theta+\sin^2\theta))$$

$$=\int_0^{\frac{\pi}{2}}\int_0^1 \sin(r^2)\,rdrd\theta$$

$r^2=t$로 치환 $\Rightarrow 2rdr=1dt \Rightarrow rdr=\frac{1}{2}dt$

$r=0\Rightarrow t=0,\ \ r=1\Rightarrow t=1$



$$=\int_0^{\frac{\pi}{2}}\int_0^1 \sin t\,\frac{1}{2}dtd\theta=\int_0^{\frac{\pi}{2}}\left[-\frac{1}{2}cost\right]_0^1 d\theta$$

$$=\int_0^{\frac{\pi}{2}}\left(-\frac{1}{2}cos1+\frac{1}{2}\right)d\theta=\left[-\frac{1}{2}(\cos1)\theta+\frac{1}{2}\theta\right]_0^{\frac{\pi}{2}}$$

$$=\frac{\pi}{4}(1-\cos1)$$

연습문제 14.3

1. 주어진 표면의 표면적을 구하여라.

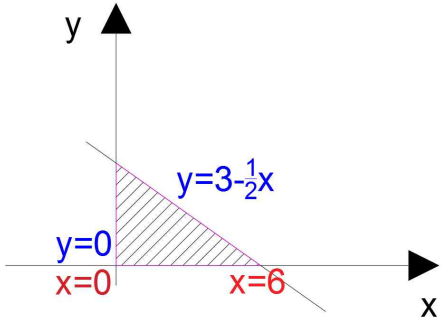
(1) 8분면 중 1분면에서의 평면 $x+2y+3z=6$의 표면적

$$S=\int_R\int \sqrt{[f_x(x,y)]^2+[f_y(x,y)]^2+1}\,dA$$

$$z=f(x,y) \Rightarrow 3z=6-x-2y \quad \therefore\ z=\frac{1}{3}(6-x-2y)$$

z와 xy평면의 교선

$$2-\frac{1}{3}x-\frac{2}{3}y=0 \Rightarrow\ y=3-\frac{1}{2}x$$



$$f_x(x,y)=-\frac{1}{3},\ f_y(x,y)=-\frac{2}{3}$$

$$S=\int_0^6\int_0^{-\frac{1}{2}x+3}\sqrt{(-\frac{1}{3})^2+(-\frac{2}{3})^2+1}\,dydx$$

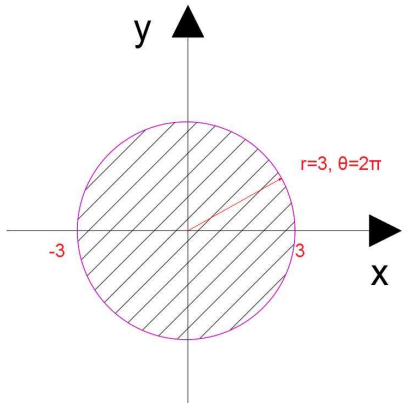
$$=\int_0^6\left[\frac{\sqrt{14}}{3}y\right]_0^{-\frac{1}{2}x+3}dx=\int_0^6\frac{\sqrt{14}}{3}(-\frac{1}{2}x+3)dx$$

$$=\frac{\sqrt{14}}{3}\left[-\frac{1}{4}x^2+3x\right]_0^6=3\sqrt{14}$$

(2) xy평면 위의 포물면 $z=9-x^2-y^2$의 표면적

$$S=\iint_R \sqrt{[f_x(x,y)]^2+[f_y(x,y)]^2+1}\,dA$$

z와 xy평면의 교선



$$x^2+y^2=9$$

$$f_x(x,y)=2x,\ f_y(x,y)=-2y$$

$$S=\iint_R \sqrt{(2x)^2+(-2y)^2+1}\,dA$$

$$=\int_0^{2\pi}\int_0^3 \sqrt{4r^2+1}\,r\,dr\,d\theta$$

$4r^2+1=t$로 치환 $8r\,dr=dt \Rightarrow r\,dr=\frac{1}{8}dt$

$r=0 \Rightarrow t=1,\ \ r=3 \Rightarrow t=37$

$$=\int_0^{2\pi}\int_1^{37}\sqrt{t}\,\frac{1}{8}\,dt\,d\theta$$

$$=\frac{1}{8}\int_0^{2\pi}\left[\frac{2}{3}t^{\frac{3}{2}}\right]_1^{37}d\theta=\frac{1}{12}\int_0^{2\pi}(37\sqrt{37}-1)\,d\theta$$

$$=\frac{1}{12}\{(37\sqrt{37}-1)\theta\}_0^{2\pi}=\frac{\pi}{6}(37\sqrt{37}-1)$$